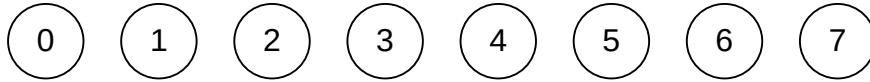
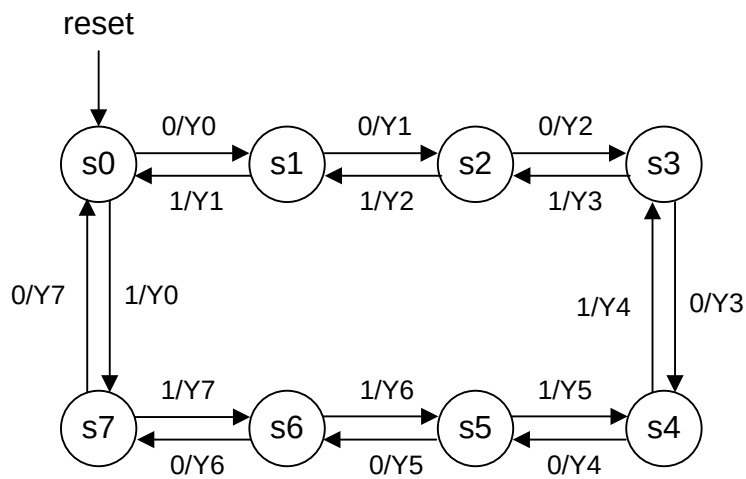


Lab 3 룰렛

8개의 LED가 1초 간격으로 차례 차례 순차적으로 점멸하며 슬라이드 스위치의 입력에 따라 점멸 방향이 바뀐다. 초기상태는 0번 LED가 점등된 상태이며 0번 LED부터 점멸을 시작한다. **reset** 버튼을 누르면 초기 상태로 되돌아 간다.



슬라이드 스위치 입력을 x 라고 하고 LED i 의 출력을 Y_i 라고 하면 FSM은 다음과 같다.



실험 키트의 보드는 100Mhz 클럭을 사용하지만 룰렛에는 1Hz 클럭을 사용한다.
1Hz 클럭은 100 Mhz 클럭을 100M 까지 카운트하여 발생시킨다.

```
always @ (posedge clk)
begin
    if (rst == 1) begin
        count <= 0;
        clkone <= 1'b0;
    end
    else if (count == 10**8) begin    // 시뮬레이션시 4
        count <= 0;
        clkone <= 1'b1;
    end
    else begin
        count <= count + 1;
        clkone <= 1'b0;
    end
end
end
```

roulette.v 소스

```
module roulette(clk,rst,x,y
    );
input clk, rst,x;
output [7:0] y;

reg clkone;
reg [26:0] count;
reg [7:0] y;
localparam s0=0, s1=1, s2=2, s3=3, s4=4, s5=5, s6=6, s7=7;
reg [2:0] state, nstate;

always @ (posedge clk)
begin
    if (rst == 1) begin
        count <= 0;
        clkone <= 1'b0;
    end
    else if (count == 10**8) begin // simulation ㄱ 4
        count <= 0;
        clkone <= 1'b1;
    end
    else begin
        count <= count + 1;
        clkone <= 1'b0;
    end
end

always @ (posedge clkone, posedge rst) begin
    if (rst == 1)
        state <= s0;
    else
        state <= nstate;
end
```

```

always @ (x, state) begin
    y = 0;

    case (state)
    s0: begin
        if (x==0) nstate = s1;
        else nstate = s7;
        y[0] = 1'b1;
    end
    s1: begin
        if (x==0) nstate = s2;
        else nstate = s0;
        y[1] = 1'b1;
    end
    s2: begin
        if (x==0) nstate = s3;
        else nstate = s1;
        y[2] = 1'b1;
    end
    s3: begin
        if (x==0) nstate = s4;
        else nstate = s2;
        y[3] = 1'b1;
    end
    s4: begin
        if (x==0) nstate = s5;
        else nstate = s3;
        y[4] = 1'b1;
    end
    s5: begin
        if (x==0) nstate = s6;
        else nstate = s4;
        y[5] = 1'b1;
    end
    s6: begin
        if (x==0) nstate = s7;

```

```

        else nstate = s5;
        y[6] = 1'b1;
    end
    default: begin
        if (x==0) nstate = s0;
        else nstate = s6;
        y[7] = 1'b1;
    end
endcase

// synthesis translate_off
    $display("state = %d, y = %b", nstate, y);
// synthesis translate_on

end

endmodule

```

tb.v test bench file

```
module tb;

    // Inputs
    reg clk;
    reg rst;
    reg x;

    // Outputs
    wire [7:0] y;

    // Instantiate the Unit Under Test (UUT)
    roulette uut (
        .clk(clk),
        .rst(rst),
        .x(x),
        .y(y)
    );

    // clock generation
    initial begin
        clk = 0;
        forever #5 clk = ~clk;
    end

    // reset
    initial begin
        rst = 0;
        #100 rst = 1;
        #100 rst = 0;
    end
end
```

```
// Initialize Inputs
initial begin
    x = 0;
    #2000 x = 1;
    #2000;
end

endmodule
```

roulette.xdc user constraint file

```
set_property PACKAGE_PIN E3 [get_ports clk]
set_property IOSTANDARD LVCMOS33 [get_ports clk]
create_clock -add -name sys_clk_pin -period 10.00 -waveform {0 5} [get_ports clk]
```

```
set_property PACKAGE_PIN N17 [get_ports rst]
set_property IOSTANDARD LVCMOS33 [get_ports rst]
```

```
set_property PACKAGE_PIN J15 [get_ports x]
set_property IOSTANDARD LVCMOS33 [get_ports x]
```

```
set_property PACKAGE_PIN H17 [get_ports y[0]]
set_property IOSTANDARD LVCMOS33 [get_ports y[0]]
```

```
set_property PACKAGE_PIN K15 [get_ports y[1]]
set_property IOSTANDARD LVCMOS33 [get_ports y[1]]
```

```
set_property PACKAGE_PIN J13 [get_ports y[2]]
set_property IOSTANDARD LVCMOS33 [get_ports y[2]]
```

```
set_property PACKAGE_PIN N14 [get_ports y[3]]
set_property IOSTANDARD LVCMOS33 [get_ports y[3]]
```

```
set_property PACKAGE_PIN R18 [get_ports y[4]]
set_property IOSTANDARD LVCMOS33 [get_ports y[4]]
```

```
set_property PACKAGE_PIN V17 [get_ports y[5]]
set_property IOSTANDARD LVCMOS33 [get_ports y[5]]
```

```
set_property PACKAGE_PIN U17 [get_ports y[6]]
set_property IOSTANDARD LVCMOS33 [get_ports y[6]]
```

```
set_property PACKAGE_PIN U16 [get_ports y[7]]
set_property IOSTANDARD LVCMOS33 [get_ports y[7]]
```